

# Explain nonnegative data with additive latent factors

Nonnegative Matrix Factorization (NMF) explains nonnegative data with additive latent factors and can support interpretable parts.

## Central idea

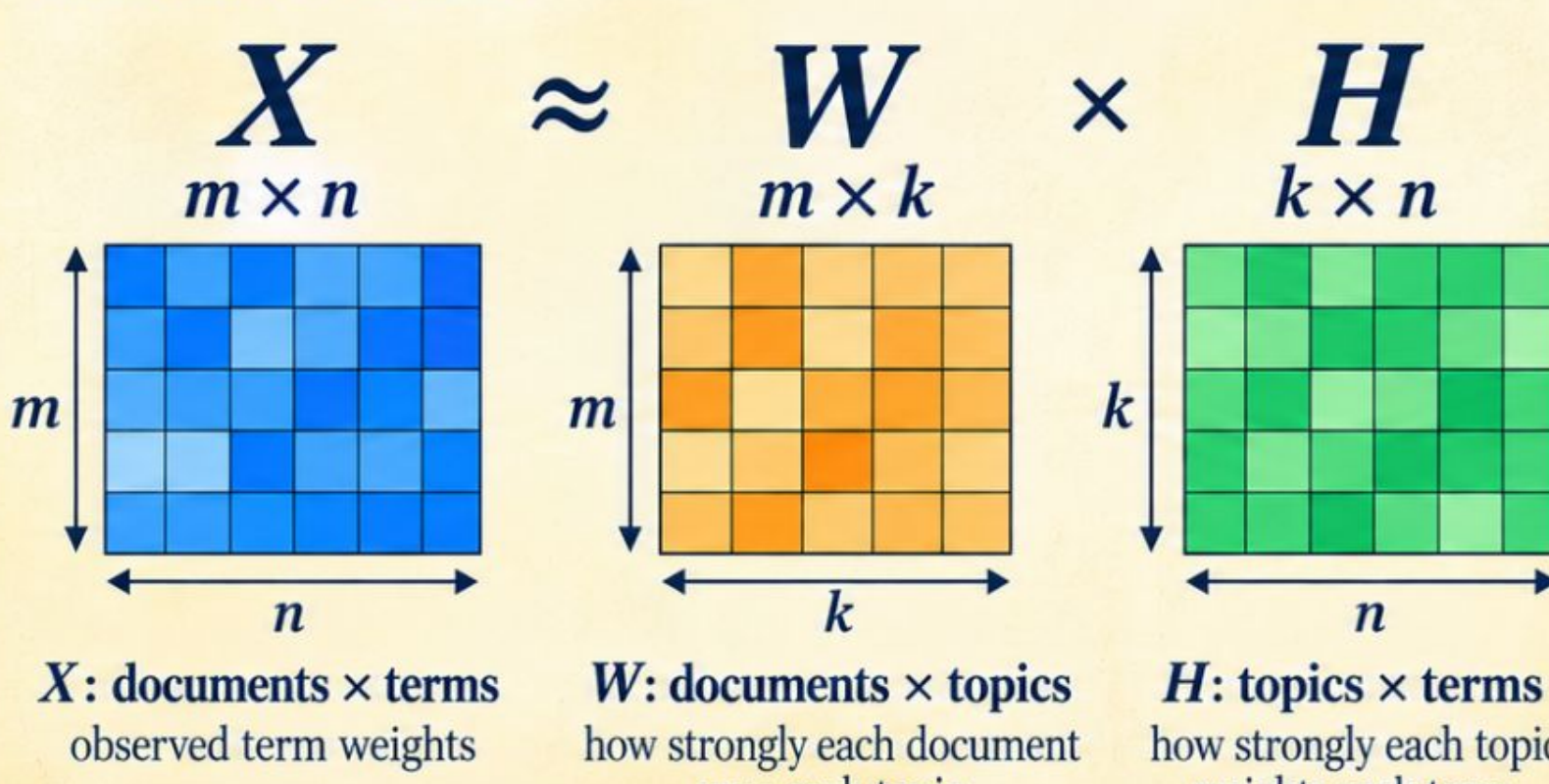
NMF explains nonnegative data through additive latent factors. Instead of allowing positive and negative parts to cancel, it builds each observation from nonnegative combinations of reusable components and can support interpretable parts.

## Running example: scientific documents

Assume a corpus of scientific documents represented as a document-term matrix. One learned topic might be "geology", with high-weight terms such as "basalt", "strata", "ridge", and "stone".

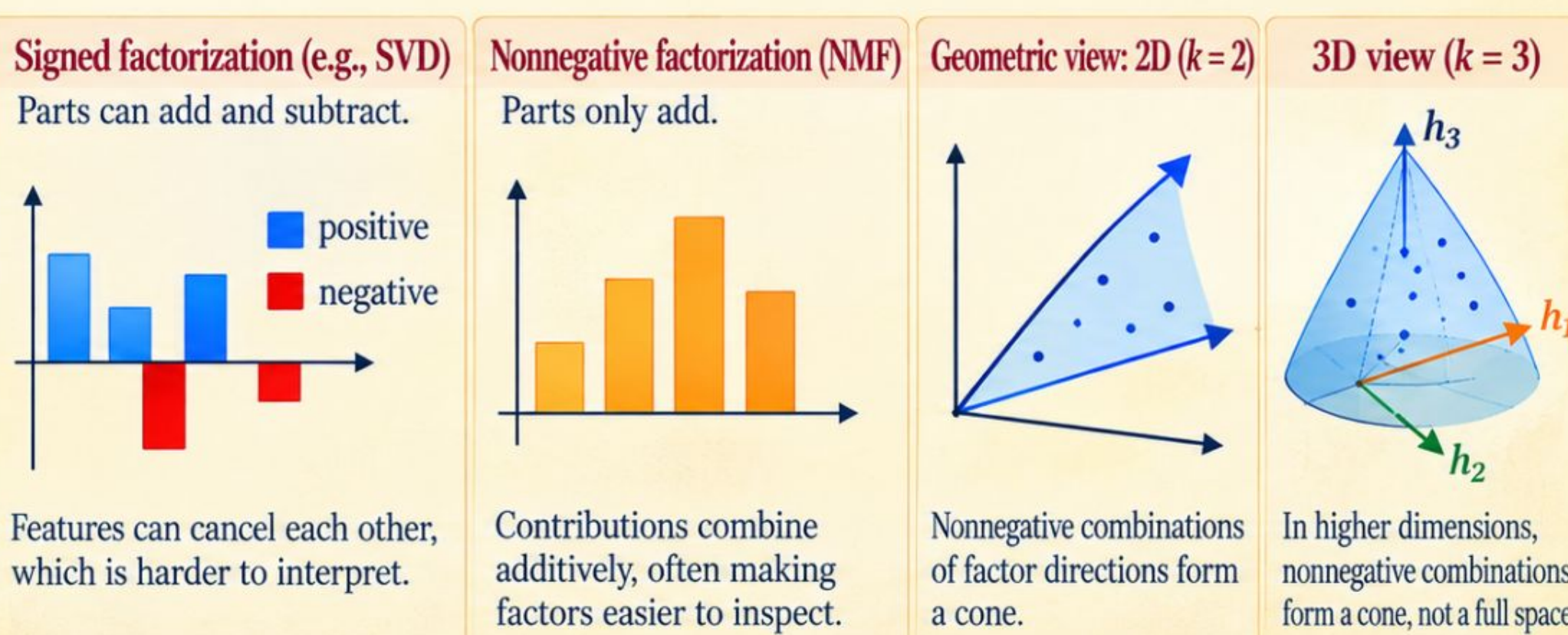
## 1 What is NMF?

For our topic-modeling example, NMF approximates a nonnegative document-term matrix  $X$  with two nonnegative matrices  $W$  and  $H$ .



## 2 Why require nonnegative factors?

Restricting contributions to nonnegative values removes cancellation and encourages an additive, parts-based representation.



## 3 Where is NMF useful? Three examples with the same structure

### Topic modeling (text)

A document can be represented by nonnegative word weights. NMF learns recurring topics from the data.



|          |                                                                                                        |
|----------|--------------------------------------------------------------------------------------------------------|
| Data     | Documents $\times$ terms<br>(e.g., TF-IDF)                                                             |
| Learns   | $W$ = document-topic weights<br>$H$ = topic-term weights                                               |
| Why NMF? | Each document becomes a mixture of recurring topics, and each topic becomes a weighted group of words. |

### Hyperspectral imaging (remote sensing)

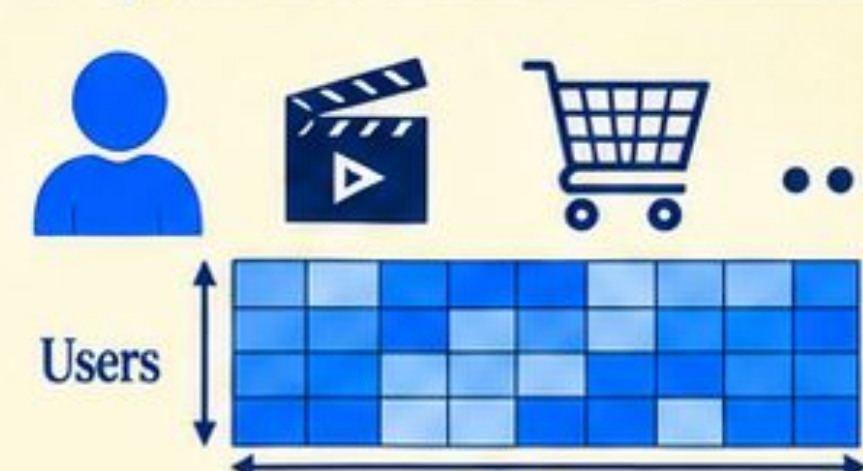
An ordinary RGB camera records three broad color channels. A hyperspectral sensor records many narrow wavelength bands, helping scientists distinguish materials that may look similar in ordinary imagery. One pixel represents an area of ground, not an infinitesimal point.



|          |                                                                            |
|----------|----------------------------------------------------------------------------|
| Data     | Pixels $\times$ wavelength bands                                           |
| Learns   | $W$ = pixel-material mixture weights<br>$H$ = material spectral signatures |
| Why NMF? | A pixel spectrum can be an additive mixture of materials.                  |

### Recommendation systems (user-item data)

A user's preferences combine several latent tastes.



|          |                                                                       |
|----------|-----------------------------------------------------------------------|
| Data     | Users $\times$ items                                                  |
| Learns   | $W$ = user-factor affinities<br>$H$ = factor-item affinities          |
| Why NMF? | Recommendations can be modeled as additive mixtures of latent tastes. |

## 4 How does NMF work (for topic modeling)?

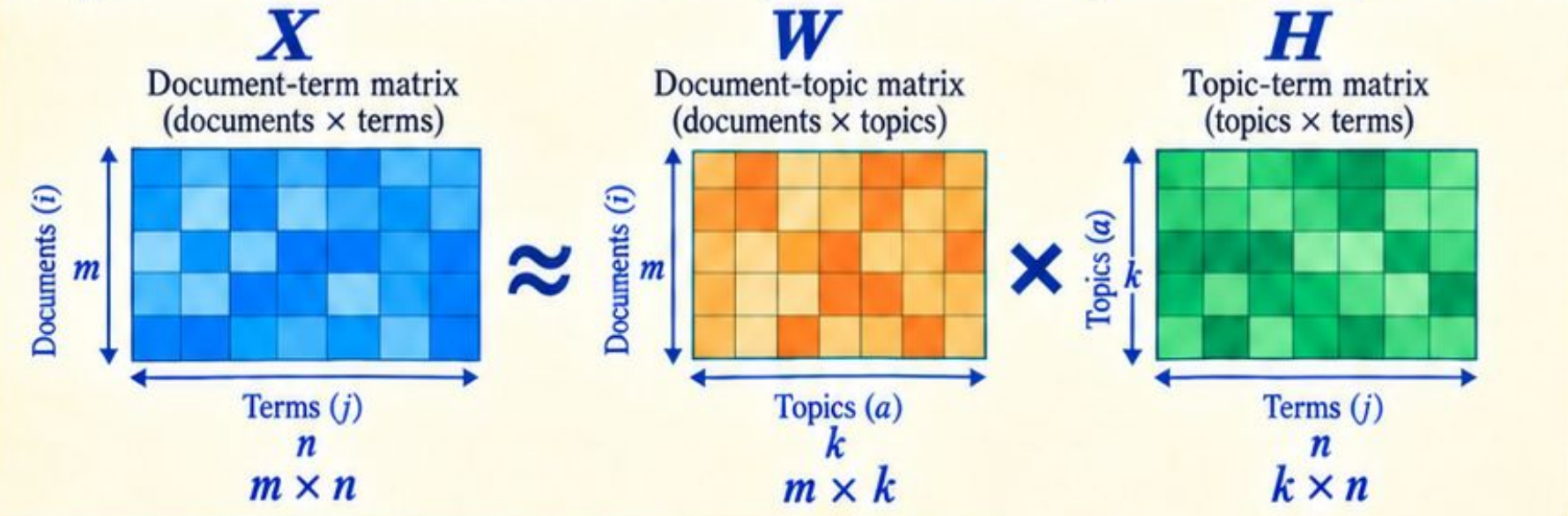
A topic-modeling view of alternating optimization.

### Building on the running example (scientific documents)

From the scientific corpus introduced earlier, represent each document with nonnegative term weights (e.g., TF-IDF). NMF asks whether those observed document-term weights can be reconstructed from a smaller set of latent topics.

### Topic-modeling view of the factorization

We approximate the document-term matrix  $X$  with the product of a document-topic matrix  $W$  and a topic-term matrix  $H$ .

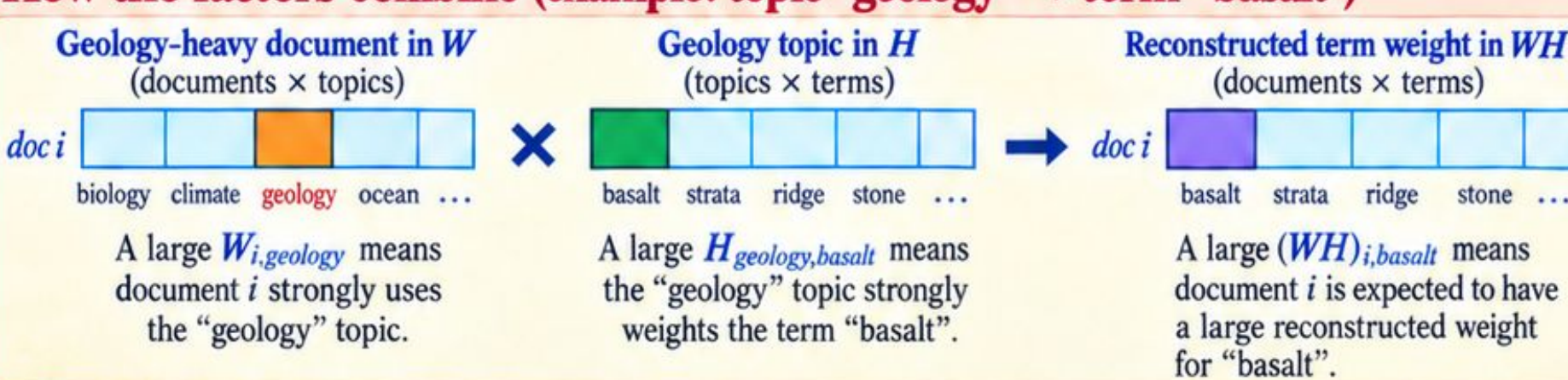


**$X$  • What we observe**  
Each row is a document; each column is a term (e.g., TF-IDF weights).

**$W$  • Document-topic weights**  
 $W_{ia}$  tells how strongly document  $i$  uses topic  $a$ . A geology-heavy document receives a large weight for the geology topic.

**$H$  • Topic-term weights**  
 $H_{aj}$  tells how strongly topic  $a$  weights term  $j$ . The geology topic might give large weights to "basalt", "strata", "ridge", and "stone".

### How the factors combine (example: topic "geology" → term "basalt")



Each document becomes an additive mixture of topics; each topic becomes a weighted group of words.

$WH$  reconstructs the document-term weights we expect to see in each document.

### NMF algorithm (alternating optimization)



### Programming view (scikit-learn)

```
from sklearn.decomposition import NMF
nmf = NMF(
    n_components=k,  # number of topics (k)
    init='nndsvd',  # initialization method
    max_iter=500,    # iteration cap
    random_state=42, # reproducibility seed
)
W = nmf.fit_transform(X) # document-topic matrix (m x k)
H = nmf.components_      # topic-term matrix (k x n)
```

### Key arguments

| Parameter        | What it does                                       |
|------------------|----------------------------------------------------|
| n_components     | number of topics $k$                               |
| init             | starting method for $W, H$ (e.g., 'nndsvd')        |
| max_iter         | iteration cap                                      |
| random_state     | reproducibility seed                               |
| fit_transform(X) | returns document-topic matrix $W$ ( $m \times k$ ) |
| components_      | returns topic-term matrix $H$ ( $k \times n$ )     |

## 5 What are we optimizing?

Minimize the reconstruction loss.

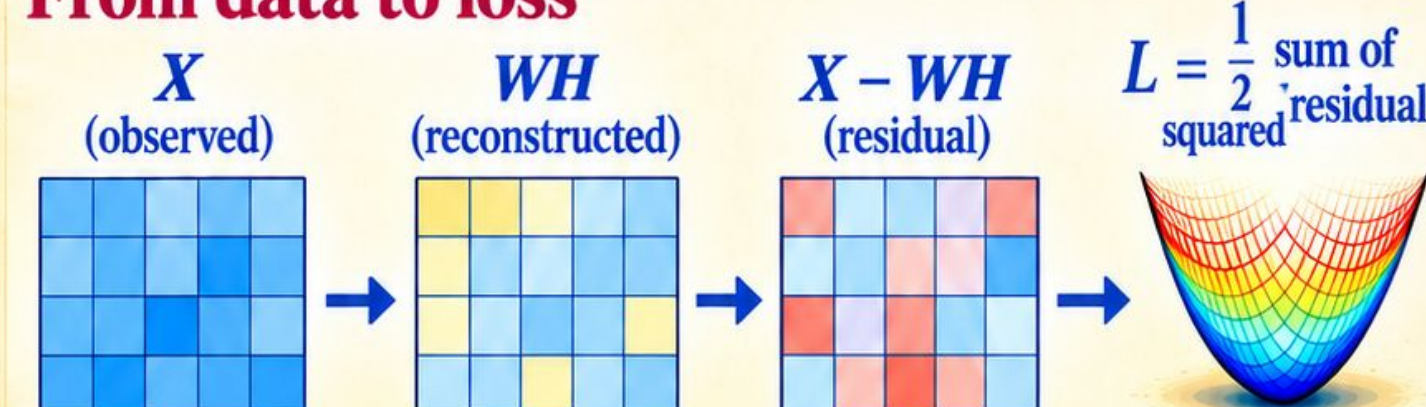
### Main objective (Frobenius loss)

$$L = \frac{1}{2} \|X - WH\|_F^2 = \frac{1}{2} \sum_{i,j} (X_{ij} - (WH)_{ij})^2$$

### Read it as:

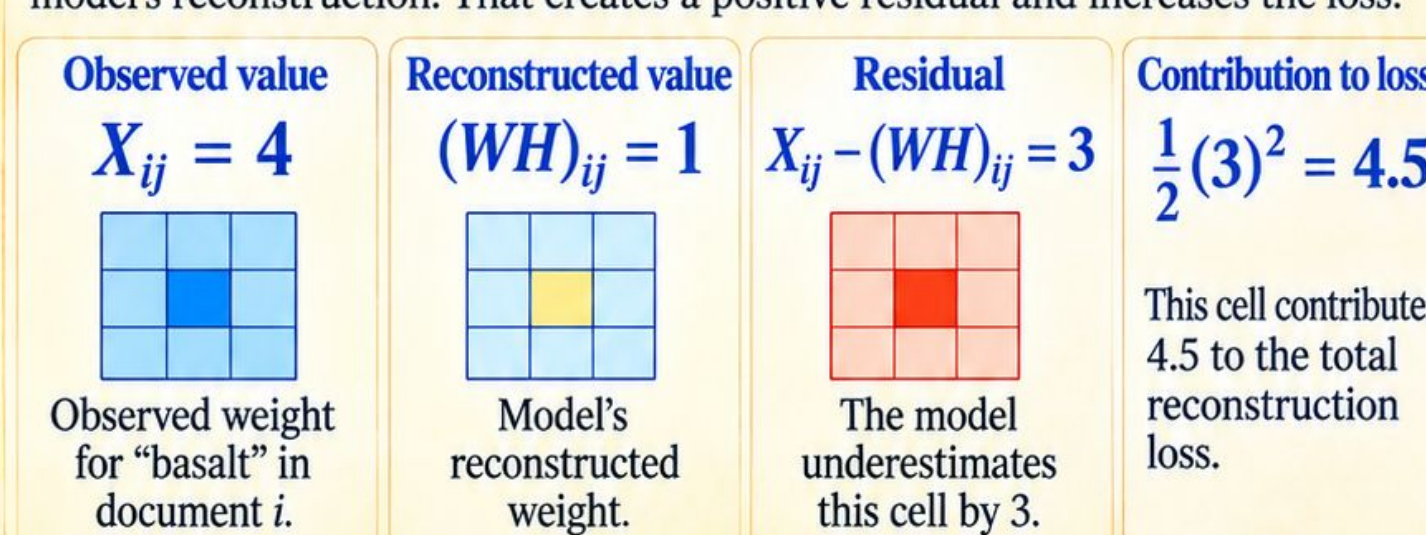
- Find nonnegative  $W$  and  $H$  whose product  $WH$  reconstructs  $X$  as closely as possible.
- Measure error by the sum of squared residuals over all cells.
- The factor  $1/2$  does not change the optimum and makes derivatives cleaner.

### From data to loss



### A single-cell example (topic "geology" / term "basalt" in one document)

In a geology-heavy document, the observed "basalt" weight is larger than the model's reconstruction. That creates a positive residual and increases the loss.



Repeat this for every cell and add the squared residuals to get the reconstruction loss:

$$L = \frac{1}{2} \|X - WH\|_F^2$$

### Error terms at a glance

| Residual (matrix form)                                                                                     | Cell-level discrepancy                                                                                                                     | Reconstruction loss                                                          | Sign convention: same mismatch, opposite sign.                                                                                                                                                                                      |
|------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $X - WH$<br>Observed minus reconstructed. Shows where the model underestimates or overestimates each cell. | $(WH)_{ij} - X_{ij}$<br>Reconstructed minus observed for one cell. This is the negative of the residual and the form used in the gradient. | $L = \frac{1}{2} \ X - WH\ _F^2$<br>Sum of squared residuals over all cells. | Residual: $X_{ij} - (WH)_{ij}$ (observed - reconstructed)<br>Cell-level discrepancy: $(WH)_{ij} - X_{ij}$ (reconstructed - observed)<br>Relationship: $(WH)_{ij} - X_{ij} = -(X_{ij} - (WH)_{ij})$<br>Same mismatch, opposite sign. |

## 6 How do we update $H$ ? From the gradient to the multiplicative update

Hold  $W$  fixed and update  $H$ . The same logic (with roles reversed) updates  $W$ .

**Notation**

$i$  = document index ( $1 \dots m$ )  
 $j$  = term index ( $1 \dots n$ )  
 $a$  = topic index ( $1 \dots k$ )

$X_{ij}$  = observed term weight  
 $W_{ia}$  = topic weight in document  
 $H_{aj}$  = term weight in topic

**Solver guardrail:** Alternating optimization is the general NMF pattern; the Lee-Seung multiplicative updates below are one classic way to perform those updates.

### 6.1 Take the derivative for one element $H_{aj}$

Differentiate the loss with respect to a single topic-term weight.

$$\frac{\partial L}{\partial H_{aj}} = \sum_i W_{ia} \left( (WH)_{ij} - X_{ij} \right)$$

Cell-level discrepancy

### Read it as:

- For term  $j$ , sum across all documents  $i$ .
- Multiply each document's topic weight  $W_{ia}$  by the cell-level discrepancy  $(WH)_{ij} - X_{ij}$ .
- If the model overestimates a cell, the discrepancy is positive; if it underestimates, the discrepancy is negative.

**Example: topic "geology", term "basalt"**  
If geology-heavy documents (large  $W_{ia}$ ) contain more observed "basalt" than the model expects, the negative discrepancy drives a larger increase in the weight.

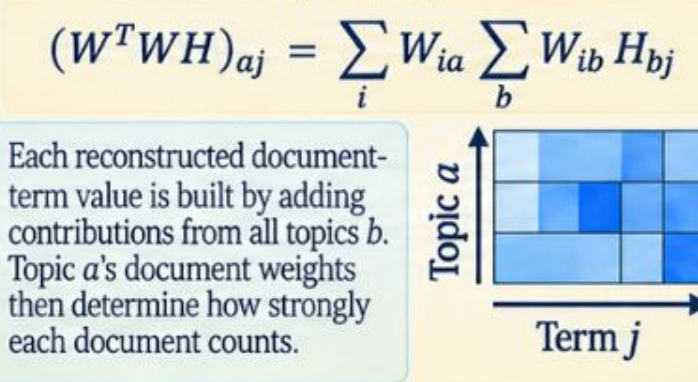
### 6.2 Assemble the full gradient for all $H$ elements

Collect those entrywise derivatives into the full matrix gradient.

$$\nabla_H L = \underbrace{W^T WH}_{\text{Reconstructed support (D)}} - \underbrace{W^T X}_{\text{Observed support (N)}}$$

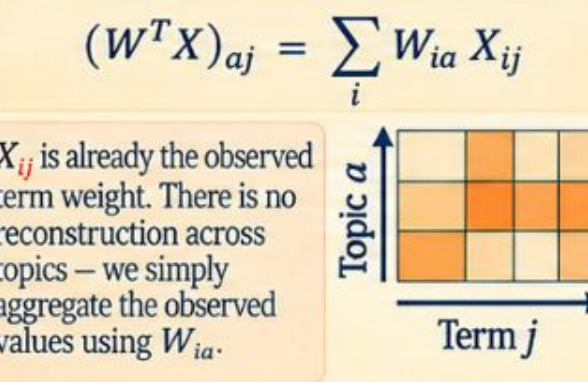
### Reconstructed support $(W^T WH)_{aj}$

Across documents  $i$ , sum over latent topics  $b$ . Each document's contribution to term  $j$  is the model reconstruction  $\sum_b W_{ib} H_{bj}$ , weighted by how strongly that document uses topic  $a$  ( $W_{ia}$ ).



### Observed support $(W^T X)_{aj}$

Across documents  $i$ , use the observed term weight  $X_{ij}$  directly, weighted by how strongly document  $i$  uses topic  $a$  ( $W_{ia}$ ).



### Read it as:

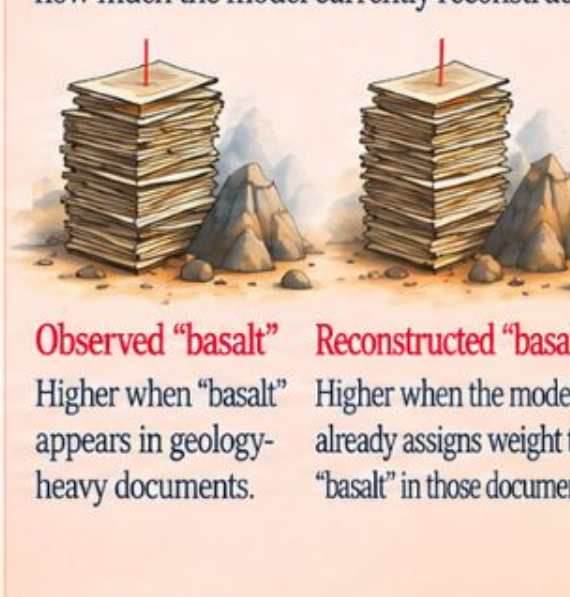
- For each topic-term pair  $(a, j)$ , compare the model's reconstructed support  $(W^T WH)_{aj}$  with the observed support  $(W^T X)_{aj}$  across all documents.
- The gradient  $\nabla_H L$  is their difference: positive where the model overestimates, negative where it underestimates.

### Why $W^T$ ?

In 6.1 we have  $W_{ia}$ , the topic weight for a single document  $i$ . The gradient sums over all documents, and  $W^T$  puts topics ( $a$ ) on rows, so document-level values aggregate into topic  $\times$  term totals.

### Example: topic "geology", term "basalt"

$(W^T X)_{aj}$  measures how much "basalt" actually appears in geology-weighted documents, while  $(W^T WH)_{aj}$  measures how much the model currently reconstructs.



### 6.3 Choose an entry-specific step size

Use a scaled gradient step with a special learning rate.

$$H_{aj}^{new} = H_{aj} - \eta_{aj} (D - N)$$

$D = (W^T WH)_{aj}$     $N = (W^T X)_{aj}$

**Choose**  $\eta_{aj} = \frac{H_{aj}}{D}$

**Read it as:**

- Scale the step size for each entry by its current value and the reconstructed support.
- This leads to a nonnegative multiplicative update while preserving nonnegativity.

**Example: topic "geology", term "basalt"**

- If the observed "basalt" support is larger than the reconstructed support ( $N > D$ ), then  $N/D > 1$ , so the weight increases.
- If the model overpredicts "basalt" ( $N < D$ ), the weight decreases.

### 6.4 Substitute and simplify to get the multiplicative update

Plug in the step size and simplify.

$$H_{aj}^{new} = H_{aj} - \left( \frac{H_{aj}}{D} \right) (D - N)$$

$$= H_{aj} - H_{aj} + H_{aj} \frac{N}{D}$$

$$= H_{aj} \frac{N}{D}$$

**Example: topic "geology", term "basalt"**

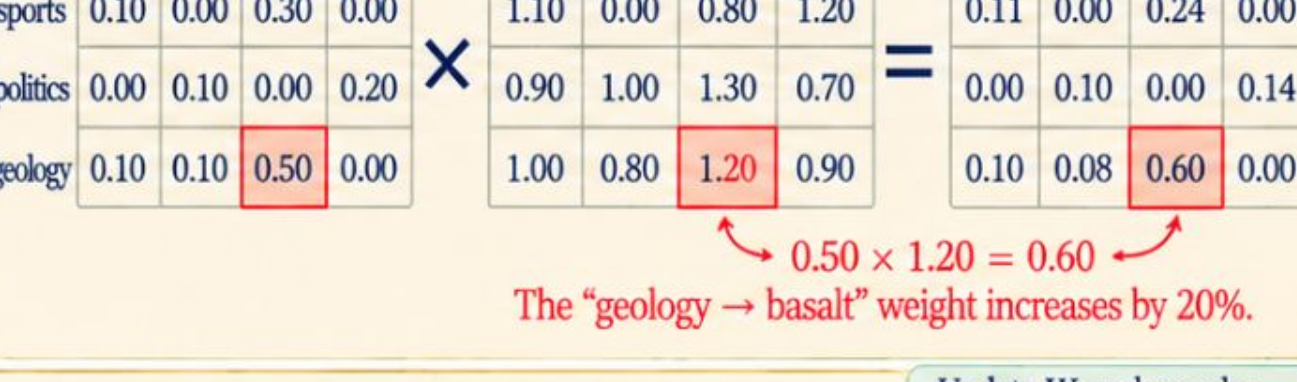
- If  $N/D = 1.2$ , the "geology → basalt" weight increases by 20%.

### 6.5 Apply the update to the whole matrix $H$

Update every topic-term weight at once using elementwise multiplication.

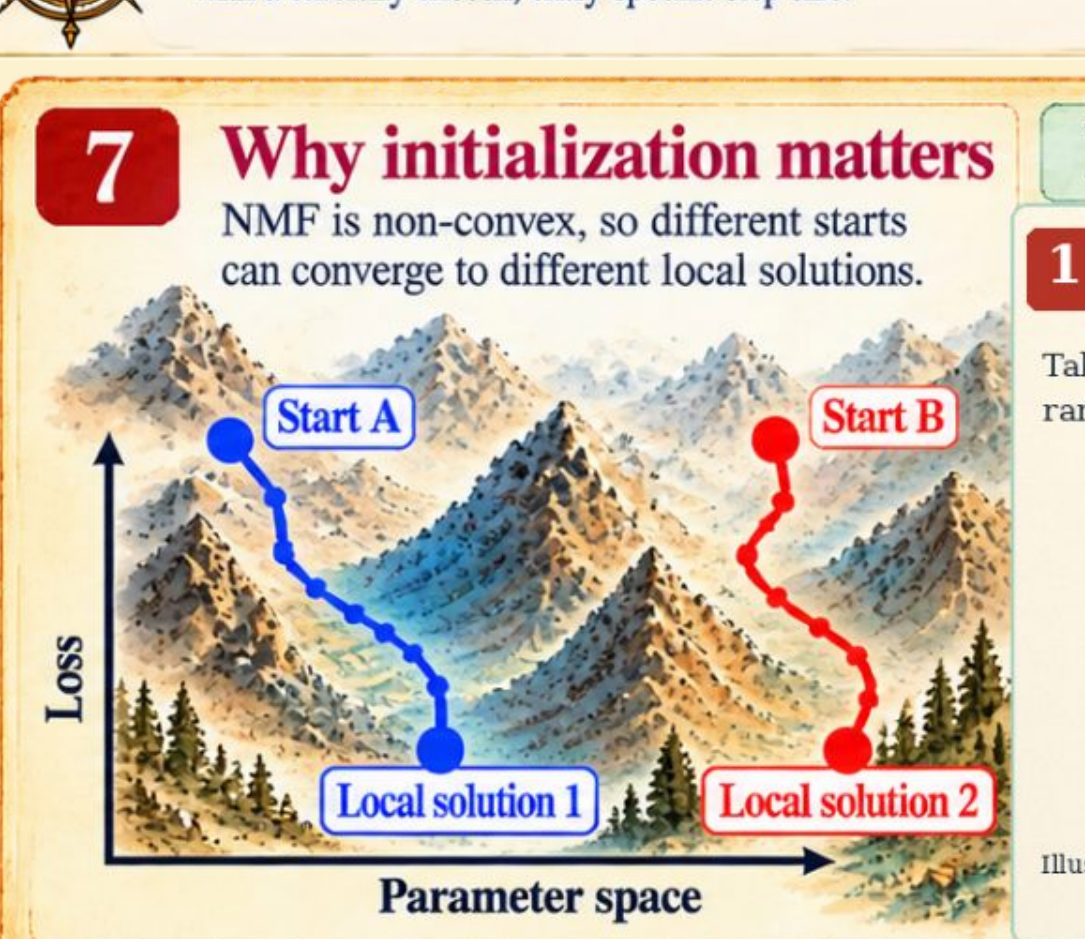
$$H^{new} = H \odot R \quad R = \frac{W^T X}{W^T WH}$$

The division in  $R$  is elementwise (each entry  $N/D$ ).



## 7 Why initialization matters

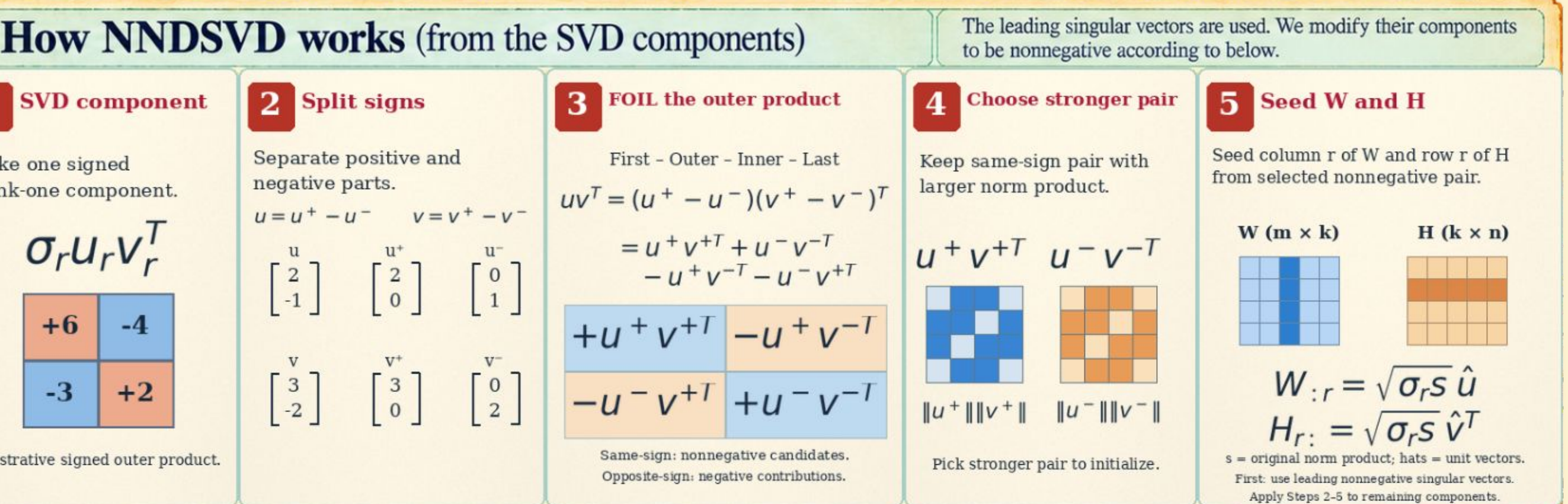
NMF is non-convex, so different starts can converge to different local solutions.



### Initialization options (scikit-learn)

```
NMF(n_components=k, init='nndsvd', max_iter=500, random_state=42)
```

### How NNDSD works (from the SVD components)



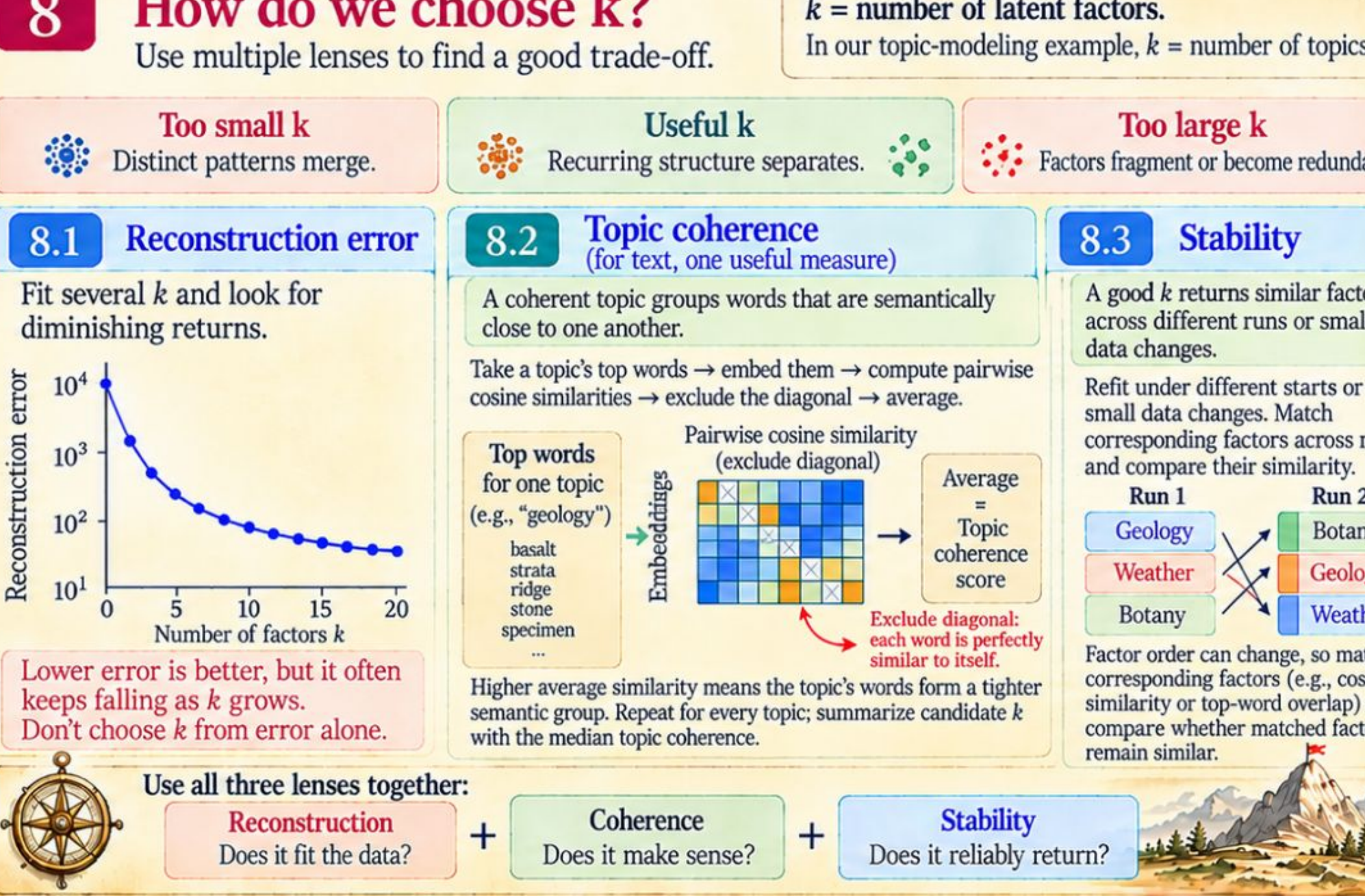
**Keep in mind**

- NNDSD is an initialization, not the final NMF solution.
- The sign-splitting step does not preserve the SVD orthogonality.

## 8 How do we choose $k$ ?

Use multiple lenses to find a good trade-off.

$k$  = number of latent factors.  
In our topic-modeling example,  $k$  = number of topics.



## 9 When would I use NMF, SVD, LDA, or transformer-based methods?

| Method            | Representation                                                                                         | Main trade-off                                                               | Scenario / goal                                                                                    |
|-------------------|--------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| SVD / LSI         | Signed low-rank factors ( $U, \Sigma, V^T$ )                                                           | Compact and algebraically useful; less directly inspectable.                 | When you need compact representations for similarity/retrieval; interpretability is not essential. |
| NMF               | Nonnegative observation-factor and factor-feature weights ( $W, H$ ).                                  | Additive and inspectable; non-convex, not probabilistic.                     | When you want recurring themes and the exact words/documents behind them.                          |
| LDA               | Document-topic and topic-word probability distributions ( $\theta, \phi$ ).                            | Explicit probabilistic meaning; stronger modeling assumptions and inference. | When probabilistic topic proportions matter — e.g., tracking topic prevalence over time.           |
| Transformer-based | Contextual embeddings plus downstream grouping (e.g., clustering documents or embeddings into topics). | Richer semantics; less feature-level transparency and more compute.          | When context, synonymy, or polysemy matter more than shared vocabulary alone.                      |

**Do not ask which method is best.**  
Ask what meaning the representation must preserve.

## 10 Why identifiability depends on the data

Distinct factors are easier to identify when the data vary enough to reveal them.

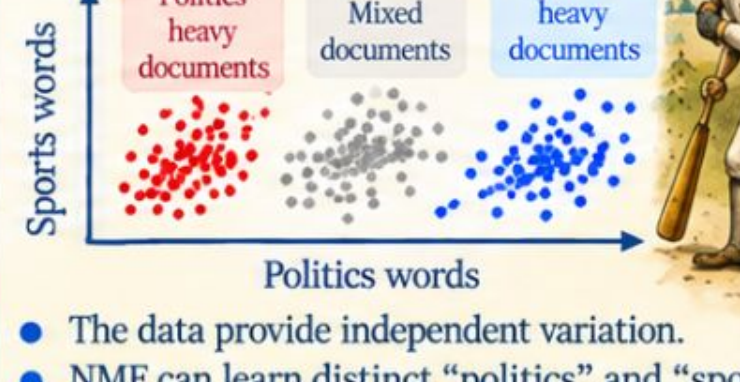
### Hard cases: factors always co-occur

If two topics always appear together, NMF may recover a single factor that combines them, even if we would like separate factors.



### Easier case: corpus exposes the parts

If the corpus contains politics-heavy, sports-heavy, and mixed documents, NMF can more easily recover separate factors.



**Interpretability is not guaranteed by the method alone; the data must reveal the parts distinctly.**  
Even when factors separate, their order can be arbitrary, and reciprocal rescaling of  $W$  and  $H$  leaves  $WH$  unchanged.

## 11 Computational complexity

Exact (general) NMF is NP-hard, which means no polynomial-time algorithm is known to be guaranteed to find the global optimum in the general case.

| Class       | Meaning (informal definition)                                                           |
|-------------|-----------------------------------------------------------------------------------------|
| P           | Problems solvable in polynomial time.                                                   |
| NP          | A proposed solution can be checked in polynomial time.                                  |
| NP-hard     | At least as hard as the hardest problems in NP (every NP problem can be reduced to it). |
| NP-complete | Both NP and NP-hard.                                                                    |

### Practical implication

- Non-convexity → different starts can reach different local optima.
  - NP-hardness → exact global optimization is computationally hard in general.
- Practical NMF therefore relies on iterative local optimization, initialization strategies such as NNDSD, and stopping criteria to find useful solutions in practice.
- NP-hard does not mean every practical NMF run is slow or exponential. It means there is no known efficient algorithm guaranteed to solve the general exact problem globally.

## 12 Remember

- NMF explains nonnegative data through additive latent factors. Reconstruction comes from adding nonnegative factor contributions.
- The factors are learned, not simply revealed. Results depend on the data,  $k$ , initialization, and optimization.
- Interpretability is earned, not guaranteed. Good factors reconstruct well, make sense, remain stable, and are supported by data that expose the parts.